Probability & Combinatorics: Permutations & Combinations

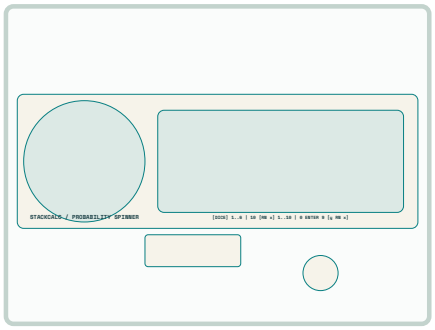
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.HSS.CP.B.9

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Probability Spinner

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Spin the dual probability spinner; roll fair dice pack to record empirical sample spaces.

Sketch Model on Paper:
Draw a tree diagram showing branch choices and outcome paths.

3. Step-by-Step LCD Stack Trace (SC-32)

MODE:	123	Base-10 Arithmetic	
1.	10	X: 10	Population n = 10 in
2.	ENTER	Y: 10 | X: 10	Stack lift
3.	3	Y: 10 | X: 3	Sample size r = 3 in
4.	nCr	X: 120	Combinations: 10C3 =

EXPECTED LCD X: 120 combinations

1 **Calculate 9C3 (choose 3 items from 9).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

2 **Calculate 6P4 (permutations: order matters).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

3 **Calculate 5! (5 factorial) on SC-32: [5] [x!].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

4 **Find probability of rolling a sum of 7 on 2 dice: 6/36.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

Station Diagnostic & Solution Key: AP Stats Unit 4

Probability & Combinatorics: Permutations & Combinations • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Probability Spinner & Fair Dice Pack

CAD Blueprint: probability_spinner

Target Device: SC-32 Scientific

CAD / STL Source File:

Education/scad/lib/probability_spinner.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal Spinner Housing with Yellow Indicator Rotor (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Calculate 9C3 (choose 3 items from 9).** Standard: CCSS.MATH.HSS.CP.B.9	84	[9] [ENTER] [3] [nCr]
2	**Calculate 6P4 (permutations: order matters).** Standard: CCSS.MATH.HSS.CP.B.9	360	[6] [ENTER] [4] [nPr]
3	**Calculate 5! (5 factorial) on SC-32: [5] [x!** Standard: CCSS.MATH.HSS.CP.B.9	120	[5] [x!]
4	**Find probability of rolling a sum of 7 on 2** Standard: CCSS.MATH.HSS.CP.B.9	0.1667 (1/6)	[6] [ENTER] [3][6] [÷]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Order matters in Permutations (nPr); order does NOT matter in Combinations (nCr).

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.